

Array Technologies Inc

Utility-Scale Solar Tracker Manufacturer — IRA/Tariff Crosswind

PhD Due Diligence Report | BLRI-DD-ARRAY-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

Solar Trackers as K Variable Infrastructure

Array Technologies manufactures single-axis solar trackers that increase utility-scale solar output by 20-25% vs fixed-tilt arrays. As the world's largest solar tracker manufacturer by installed GW, ARRAY is a K variable enabler — its technology improves the energy yield of the solar capital stock globally. At \$5.195 (mkt cap \$800M), ARRAY trades below book value despite the utility-scale solar buildout accelerating. The market prices in tariff uncertainty on Chinese steel and supply chain disruption risk.

Source: moomoo OpenD [2026-08-13] · Array Technologies 10-Q · IRA domestic content provisions

PART II — QUALITATIVE ANALYSIS

DuraTrack Technology: Structural Advantage in Wind and Slope Tolerance

Array's DuraTrack HZ tracker is designed for high-wind and cross-slope terrain that standard trackers cannot handle. This technical differentiation opens geographies (Texas plains, mountain-adjacent projects) inaccessible to flat-terrain competitors. The IRA domestic content adder (10% additional ITC) rewards US-manufactured trackers — Array's US manufacturing qualifies, while Chinese competitors face 201/301 tariff exposure. Key headwind: NI -\$112M reflects commodity cost pressure on steel and aluminium inputs.

Source: Array Technologies 10-Q EDGAR · SEIA utility-scale tracker market report · IRA domestic content guidance

PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]

Price: \$5.195 | Prev Close: \$5.19 | Mkt Cap: \$800.1M
EPS TTM: -\$0.73 | Net Income: -\$112.4M | Div Yield: 0%
52w: \$4.77–\$12.23 | Shares: 154.0M

Metric	Value	Source
Price (2026-08-13)	\$5.195	moomoo OpenD
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EPS TTM	-\$0.73	moomoo OpenD
Net Income TTM	-\$112.4M	moomoo OpenD
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PART IV — COMPETITIVE LANDSCAPE

Competitor	Type	Est. Share	vs ARRY
Nextracker (NXT)	Smart tracker + AI optimisation	~40% global	NXT profitable; AI differentiation
Soltec (Spain)	Bifacial single-row tracker	~12%	European focus; ARRY leads US
Arctech Solar (China)	Chinese tracker leader	~15% global	Tariff-exposed in US market
GameChange Solar	Private US tracker	~8% US	Private; ARRY listed advantage

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The Utility-Scale Solar EPC Engineer (COMPOSITE)

A project engineer at a Texas 500 MW solar farm evaluates tracker options. 'Array's DuraTrack handles the cross-slope we have in West Texas. Nextracker required 15% more civil grading work. We chose Array — \$3M saved on site prep.' Represents the EPC engineering buyer for whom terrain adaptability is a primary differentiator over software features.

COMPOSITE / ILLUSTRATIVE — drawn from utility-scale solar EPC procurement patterns

FINDING F-5-B: Portrait 2: The Solar Developer Procurement Director (COMPOSITE)

A procurement director at a Texas IPP (independent power producer) runs annual tracker RFPs across Array, Nextracker, and Soltec. 'We split orders 60/40 Array/NXT. Array wins on price in complex terrain; NXT wins on software for flat-land. No single tracker fits every project.' Represents the multi-vendor procurement strategy that limits any single tracker manufacturer from monopolising large portfolios.

COMPOSITE / ILLUSTRATIVE — drawn from IPP tracker procurement patterns

FINDING F-5-C: Portrait 3: The IRA Domestic Content Analyst (COMPOSITE)

A US Treasury IRA compliance analyst reviews Array's domestic content documentation for a 200 MW project claiming the 10% adder. 'Array's US manufacturing qualifies — their steel comes from domestic mills. Chinese tracker competitors cannot claim this adder.' Represents the regulatory moat that IRA domestic content provisions create for Array vs. Chinese tracker manufacturers.

COMPOSITE / ILLUSTRATIVE — drawn from IRA domestic content compliance patterns

PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: HH1: Nextracker's AI software creates insurmountable competitive advantage

EVIDENCE FOR:

NXT's AI-driven tracker optimisation delivers 2-5% additional energy yield vs Array; developers willing to pay premium for software edge; Array commoditised

EVIDENCE AGAINST:

Array competes on terrain adaptability and price — different buyer segments; not all projects benefit from AI optimisation (flat terrain gains are marginal)

QUANTITATIVE TEST

P(NXT wins >60% of new US tracker orders by value in 2027)=30%

VERDICT: BEAR — NXT software differentiation is real and widening; ARRY needs response

Implication: Monitor ARRY's own software roadmap announcements; NXT win rate disclosures

HH2: HH2: Steel tariff relief reduces ARRY input cost by 15%

EVIDENCE FOR:

Section 232 steel tariffs modified under bilateral trade deal; ARRY's domestic steel input cost falls; margin recovery to breakeven

EVIDENCE AGAINST:

Trade policy unpredictable; steel tariff relief would also benefit competitors

QUANTITATIVE TEST

P(meaningful steel tariff relief benefiting ARRY by 2027)=20%

VERDICT: BULL — cost structure improvement is the fastest path to ARRY profitability

Implication: Monitor Section 232 tariff review announcements; ARRY gross margin quarterly

HH3: HH3: IRA domestic content adder drives US market share gains

EVIDENCE FOR:

10% additional ITC for domestic content makes ARRY trackers preferred vs Chinese alternatives; US utility-scale solar buildout 50+ GW/year; ARRY captures 35%+ of domestic market

EVIDENCE AGAINST:

Nextracker also qualifies for domestic content; ARRY vs NXT competition intensifies

QUANTITATIVE TEST

$P(\text{ARRY US market share} > 35\% \text{ in } 2027) = 25\%$

VERDICT: BULL — domestic content moat is a genuine regulatory tailwind

Implication: Monitor ARRY order book announcements; domestic content qualification disclosures

HH4: HH4: ARRY returns to profitability in FY2027

EVIDENCE FOR:

Steel/aluminium cost normalisation + price recovery from IRA demand surge; NI positive by Q2 2027; stock re-rates from \$5 to \$8-10

EVIDENCE AGAINST:

Competitive pressure from NXT keeps ARRY pricing constrained; gross margin expansion requires commodity cost relief

QUANTITATIVE TEST

$P(\text{ARRY NI positive in FY2027}) = 35\%$

VERDICT: BULL — path to profitability exists via cost normalisation

Implication: Monitor ARRY gross margin trend; steel futures; order book growth

HH5: HH5: Acquisition by solar developer or EPC company

EVIDENCE FOR:

AES, Leeward, or EPC firm acquires ARRY for vertical integration of tracker supply; premium over depressed \$800M market cap

EVIDENCE AGAINST:

Tracker manufacturing is low-margin; acquirers prefer asset-light models; antitrust concern if utility-developer acquires manufacturer

QUANTITATIVE TEST

$P(\text{ARRY acquired at } > 30\% \text{ premium within 24 months}) = 15\%$

VERDICT: BULL — depressed valuation makes ARRY an attractive M&A target

Implication: Monitor ARRY strategic review; large developer M&A activity

PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: Return to profitability + re-rating (P = 30%%)

Steel cost normalisation; NI positive FY2027; stock \$8-10

Driver: Quarterly gross margin; steel futures; order book

Falsifier: NXT market share gains prevent ARRY price recovery

SCENARIO 2: IRA domestic content wins share from Chinese trackers (P = 25%%)

ARRY captures 35%+ US market; revenue grows 20%+ YoY; stock \$7-9

Driver: IRA domestic content qualification; order announcements

Falsifier: NXT wins domestic content share; Chinese tariff exemptions granted

SCENARIO 3: Base case: near 52-week low, marginal recovery (P = 30%%)

ARRY holds \$4.5-6.5; losses narrow but not eliminated; IRA demand offsets margin compression partially

Driver: Gross margin trend; quarterly order book

Falsifier: Losses deepen; working capital stress; dilution

SCENARIO 4: Acquisition at premium (P = 12%%)

AES/Leeward/EPC acquires at \$8-10/share; 50-90% premium on depressed valuation

Driver: M&A announcements in renewable supply chain

Falsifier: Acquirers deterred by ARRY's losses; tracker margins too thin

SCENARIO 5: NXT dominates; ARRY shrinks to niche (P = 15%%)

NXT wins 60%+ of flat-terrain US projects; ARRY relegated to complex terrain niche; stock below \$4

Driver: NXT win rate; ARRY backlog growth vs decline

Falsifier: Terrain-differentiated projects remain large enough for ARRY profitability

SCENARIO 6: Steel cost spike — losses deepen (P = 10%%)

Section 232 tariffs expand; steel costs rise 30%+; ARRY losses widen to \$200M+; dilutive equity raise below \$4

Driver: Steel futures; tariff policy changes

Falsifier: ARRY hedges steel via long-term supply contracts; cost pass-through in contracts

SCENARIO 7: Nextracker partnership or licensing deal (P = 8%%)

ARRY licenses NXT's AI software for DuraTrack terrain products; creates hybrid terrain+AI product; margin improvement

Driver: Partnership announcements; NXT licensing activity

Falsifier: NXT refuses partnership with direct competitor; ARRY must build own software

SCENARIO 8: Offshore/floating solar tracker expansion (P = 10%%)

ARRY expands into floating solar trackers for reservoir/lake installations; new product line avoids direct NXT competition

Driver: ARRY product roadmap announcements; floating solar market growth

Falsifier: Floating solar is too niche; ARRY lacks marine engineering expertise

SCENARIO 9: US utility-scale solar boom drives demand surge (P = 20%%)

50+ GW/year US utility-scale solar; ARRY order book doubles; revenue \$800M+; losses eliminated by scale

Driver: SEIA US solar forecast; ARRY order book disclosures

Falsifier: NXT captures boom demand; ARRY manufacturing constrained

SCENARIO 10: Bull case: ARRY reaches \$12+ on profitability + IRA (P = 5%%)

Profitability achieved; IRA domestic content moat confirmed; stock returns to 52-week high \$12+

Driver: Multiple catalysts: profitability + IRA + steel relief

Falsifier: NXT wins; steel stays expensive; losses persist

PART VIII — SELDON VARIABLE DEEP DIVE

Variable	ARRY Impact	Direction	Magnitude
K	Tracker tech increases solar output 20-25% per installed GW	↑ Enabling	K+ meaningful
ξ	Improves economics of utility-scale solar → broader energy access	↑ Indirect	ξ+ moderate
Ω	Displaces fossil energy → climate Ω reduction	↓ Reduces	Ω-minus moderate
I	US domestic manufacturing supports industrial employment	↑ Mild	I+ marginal

PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
NXT competitive pressure	HIGH (30%)	HIGH	Terrain differentiation; domestic content moat
Steel cost escalation	MEDIUM (20%)	HIGH	Long-term supply contracts; contract pass-through clauses
Continued losses + dilution	MEDIUM (25%)	HIGH	Gross margin expansion path; IRA demand surge
IRA modification	LOW (10%)	MEDIUM	Export market (non-IRA) provides partial offset
Customer concentration	MEDIUM (20%)	MEDIUM	Top 10 customers <50% of revenue

CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: DEEP VALUE / SPECULATIVE — TERRAIN MOAT vs. NXT SOFTWARE TIDE

ARRY at \$5.195 (mkt cap \$800.1M) is the highest-risk renewable manufacturer report. Negative NI - \$112M; trades below book; terrain-differentiation moat is real but NXT's AI edge is widening. IRA domestic content is a genuine catalyst. Path to profitability requires steel cost relief + IRA demand surge. Seldon: K+ and ξ+ via solar yield improvement. HUMAN_ONLY_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,555ch.